Shape optimization using PhiFEM

Josué Daniel Díaz Avalos[1], **Raphaël Bulle**[2], Stéphane Cotin[2], Louis Ducongé[3], Michel Duprez[2], Antoine Laurain[1]

[1] Universität Duisburg-Essen
[2] Inria de l'Université de Lorraine
[3] Université de Lorraine

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Presenter: Raphaël Bulle (Inria de l'Université de Lorraine)

The level-set method for evolving interfaces was first introduced in [1] and has been applied to a variety of shape optimization problems (see e.g. [3,4,5,6,7,8]). The core idea of this method is to use a level-set description of the domain to be optimized and modify this level-set e.g. by the derivation of optimality conditions via the shape derivative of the domain [3].

Like the level-set method, the φ-FEM takes advantage of a description of the domain via a level-set. The φ-FEM is an immersed boundary finite element method (FEM) with optimal convergence rate and conditioning which does not require any non-standard quadrature rule on cut cells nor non-standard finite element shape functions [9,10]. Thus, the φ-FEM is naturally well suited to be used with the level-set method to perform shape optimization as it provides a good description of the evolving boundary without any re-meshing of the domain.

In this talk, we describe a preliminary study on a shape optimization algorithm using φ-FEM to solve the state equations. This algorithm is implemented in FEniCSx via the recent FormOpt shape optimization toolbox [8] and the phiFEM python package [11]. We investigate the benefits of using φ-FEM instead of a standard non-conforming FEM by performing several numerical comparisons on different types of shape optimization problems.

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